DO NOW — RECALL

# Specification gap and misconception check

Name: _____ Class: _____ Date: _____

**1. What does a salt add to password storage?**

**2. What is an INNER JOIN result based on?**

**3. Which OS mechanism translates pages to frames?**

**4. Full adder A=1,B=0,carry-in=0: sum and carry, and binary total.**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

A distinct random per-account value, improving resilience of password hashing.

**2. Model answer**

Matching records according to a join condition.

**3. Model answer**

Page table.

**4. Model answer**

1+0+0=1=$01_2$; sum 1, carry 0.

**Vocabulary — know these terms**

misconception — Incorrect mental model of a mechanism.

counterexample — Case disproving a general claim.

transfer — Apply understanding to new case.

diagnostic — Detect specific gaps in knowledge.





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

**Guided 1 — original CHECK**

Show method

Encoding value/count pairs may consume more bits than the original; compression success depends on data redundancy.

IMPROVE: Explain one correction to your own reasoning.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Correct 'RLE always compresses text'.**

The input has no repeated runs longer than one symbol.




WEDO · STAGE 2: The input has no repeated runs longer than one symbol. then Encoding value/count pairs may consume more bits than the original; compression success depends on data redundancy.





NEW LEARNING

# Read, annotate and explain

### Understand

When correcting an error, identify the incorrect assumption precisely; a bare True/False is weak diagnostic evidence.

### Trace

Connect each correction to a worked example: HTTPS versus phishing, paging versus remote files, key joins versus unmatched rows.

### Evaluate

Show transfer by applying the rule to a different problem without being prompted by the original misconception.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

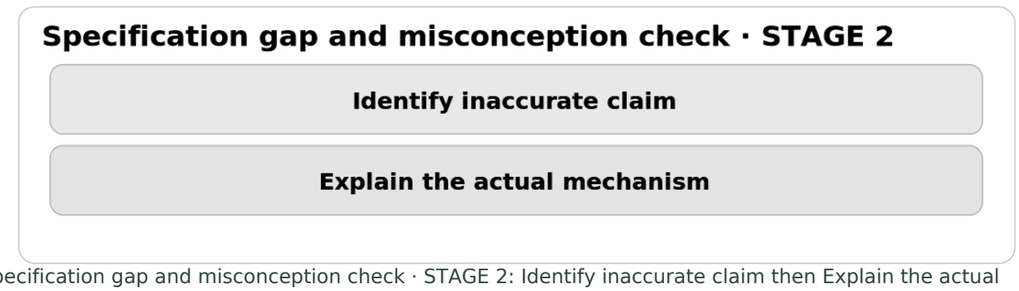
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**



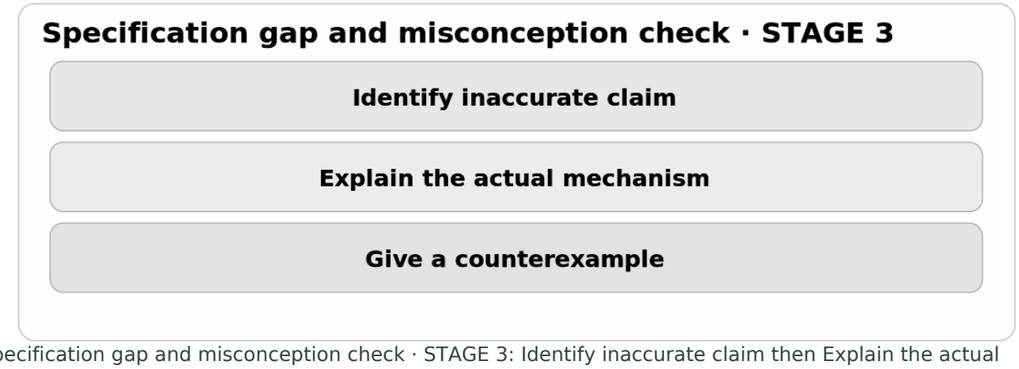

Specification gap and misconception check · STAGE 1: Identify inaccurate claim

**New Learning**




Specification gap and misconception check · STAGE 2: Identify inaccurate claim then Explain the actual mechanism

**New Learning**




Specification gap and misconception check · STAGE 3: Identify inaccurate claim then Explain the actual mechanism then Give a counterexample





**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**An accurate definition alone guarantees that you can apply a concept.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Choose three of the previous misconceptions and construct a counterexample for each. [6]

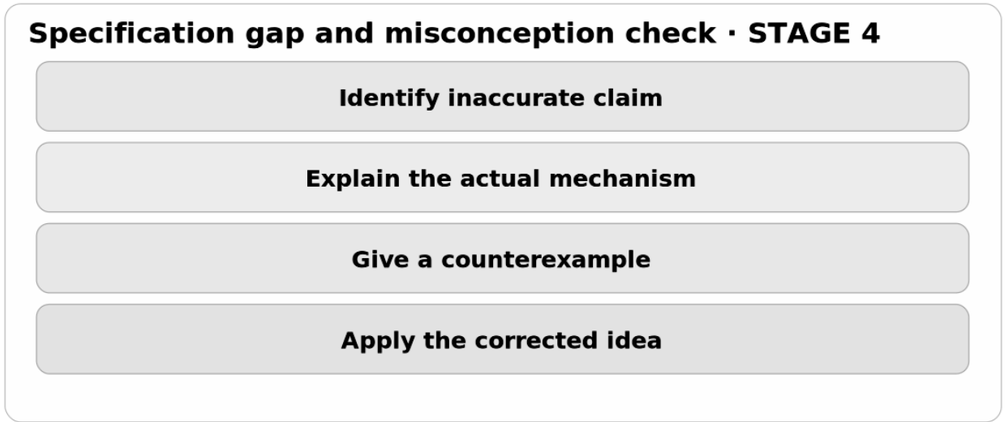




# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




Specification gap and misconception check · STAGE 4: Identify inaccurate claim then Explain the actual mechanism then Give a counterexample then Apply the corrected idea

I DO — WORKED EXAMPLE

# Follow the worked method

**Correct 'a page fault is a missing website page'.**

### Predict first

A page fault is a memory-management event where required virtual page data is not currently resident.

### Show method

HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Correct four statements: 'DNS sends HTML'; 'ROM is volatile'; 'SHA is encryption'; 'INNER JOIN keeps all rows'. [8]

### Question 2

Explain why a school's cloud files and its RAM paging system should not be described by one mechanism. [6]

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages.
DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict first The input has no repeated runs longer than one symbol.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.





# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Correct 'a page fault is a missing website page'.**

A page fault is a memory-management event where required virtual page data is not currently resident.




IDO · STAGE 1: A page fault is a memory-management event where required virtual page data is not currently resident.

**Correct 'a page fault is a missing website page'.**

A page fault is a memory-management event where required virtual page data is not currently resident.




IDO · STAGE 2: A page fault is a memory-management event where required virtual page data is not currently resident. then HTTP 404 is a web server status about a requested resource; the word 'page' refers to different concepts.

WE DO — GUIDED PRACTICE

# Try → Discuss → Improve





## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Correct 'RLE always compresses text'.**

The input has no repeated runs longer than one symbol.




WEDO · STAGE 1: The input has no repeated runs longer than one symbol.